

Joint and Osteoarthritis

Arthritis of the hip, knee and hand, joint prostheses and cartilage (GDF5, DOT1L, ALDH1A2, SMAD3), with a droplet bond

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Key variants by topic	6
5. GWAS loci burden	8
6. Favourable	9
7. Unfavourable	10
8. Mechanisms of ageing	11
9. Practical suggestions	12
10. Limitations	13
11. Methods and sources	14
12. Annex: all variants analysed	15

1. Summary

Favourable. Arthritis in general has a low to typical predisposition. The arthritis score at any location is at percentile 15.8 and the loci load at percentile 20.2. In the knee, the loci load is at 29.5 and the total joint replacement at 41.2. In the hip, the total joint replacement is at 40.6. The **GDF5 rs143383 G/G** does not have the risk allele of the highest locus of the form of the joint. RUNX2, FILIP1 and ASTN2 also do not have risk alleles. In the droplet, the genomic score is at percentile 12.3, the loci load at percentile 34.4 and the **ABCG2 Q141K** is absent (G/G), as in the §30.

Unfavourable. The hand and finger arthrosis has the loci load at percentile 97.6 (37 loci), with 1 allele in **ALDH1A2**, the strongest locus of severe hand arthrosis. **MCF2L** has 2 risk alleles and DOT1L does not have the protective allele. GLT8D1, TGFA, SLC2A9 and SLC22A12 (URAT1) have 1 allele each.

What does it mean? The genetic risk of large joint arthrosis (knee and hip) is typical or low. The genomic scores of the knee (56.1) and hip (38.9) prevail over the low coverage scores, which give 6.0 and 11.4. The exception is the hand, where genetics weigh more than in other joints. The droplet is unlikely. In arthrosis, the most important is muscle strength, absence of joint injuries and low systemic inflammation.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 12 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 12 of the chip, 0 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 1 SNPs needed conversion to the complementary strand and 2 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 10 PGS Catalog scores calculated locally with 3,130,868 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 2342 loci, 2174 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all these scores, a higher percentile = more risk. Knee scores and low coverage hip (34–44%) should be read with caution; genomics, with 97% coverage, prevail. The scores already used maintain the previous values.

Arthritis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Arthritis (any)	PGS001882 (Privé 2022, 11 451 variants)	11,157 11,451 (weighted coverage 97%)	15.8	LESS LIKELY	favourable

- **Arthritis (any):** In §27 (muscle and bone).

Knee

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Knee arthrosis (genomic)	PGS002767 (Mars 2022, PRS-CS, 1.05 million variants)	1,029,107 1,052,275 (weighted coverage 97%)	56.1	TYPICAL	typical
Knee arthrosis	PGS018416 (Wei 2024, 83 variants)	74 83 (weighted coverage 44%)	6.0	LESS LIKELY	favourable
Total knee prosthesis	PGS002749 (Lacaze 2022, 10 variants)	10 10 (weighted coverage 100%)	41.2	TYPICAL	typical

- **Knee arthrosis (genomic):** In a previous report (56.1).

Hip

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Hip arthrosis (genomic)	PGS002763 (Mars 2022, PRS-CS, 1.05 million variants)	1,029,107 1,052,273 (weighted coverage 97%)	38.9	TYPICAL	typical
Hip arthrosis	PGS018417 (Wei 2024, 70 variants)	60 70 (weighted coverage 34%)	11.4	LESS LIKELY	favourable
Total hip prosthesis	PGS002750 (Lacaze 2022, 37 variants)	34 37 (weighted coverage 69%)	40.6	TYPICAL	typical

- **Hip arthrosis (genomic):** In a previous report (38.9).

Drop

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Drop	PGS003329 (Sumpter 2022, 19 variants)	18 19 (weighted coverage 96%)	35.1	TYPICAL	typical
Drop	PGS000199 (Knevel 2020, 29 variants)	28 29 (weighted coverage 96%)	56.7	TYPICAL	typical
Gout (genomic)	PGS002762 (Mars 2022, PRS-CS, 1.09 million variants)	1,061,273 1,092,214 (weighted coverage 97%)	12.3	LESS LIKELY	favourable

- **Gout:** In §30 (Kidney and uric acid).
- **Gout:** Already in §30.

4. Key variants by topic

Replicated loci of cartilage and joint shape, hand arthrosis and urate transporters (linking to §30). The alleles with effect published in the protective sense were reversed to the sense of risk.

Cartilage and joint shape

Genes of joint development (GDF5, RUNX2) and cartilage maintenance.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
GDF5	rs143383		G/G	chip	T (0 copy/s)	favourable	5'UTR of GDF5: the low risk allele GDF5 (growth and form of the joint); ~1.2× more knee and hip arthrosis [GWAS Catalog: Knee osteoarthritis, allele T, OR 1.09, p=8E-18, PMID 30374069]	Strong
DOT1L	rs12982744		C/C	chip	C (2 copy/s)	unfavourable (light)	DOT1L (methyltransferase of histones; via cartilage Wnt): the G allele is associated with greater joint space of the hip and less arthrosis; C is the risk [GWAS Catalog: Osteoarthritis, G allele, beta + 0.09 mm of joint space]	Moderate
SUPT3H/ RUNX2	rs10948172		A/A	chip	G (0 copy/s)	favourable	RUNX2 region (condrocyte hypertrophy): arthrosis [GWAS Catalog: Osteoarthritis (hip), G allele, OR 1.09, p=9E-11, PMID 30374069]	Moderate
MCF2L	rs11842874		A/A	chip	A (2 copy/s)	unfavourable	Nucleotide exchange factor of guanine MCF2L: arthrosis [GWAS Catalog: Osteoarthritis, allele A, OR 1.17, p=2E-8, PMID 21871595]	Moderate
GLT8D1	rs6976		C/T	chip	T (1 copy/s)	unfavourable (light)	Locus GLT8D1/GNL3 (3p21): arthrosis [GWAS Catalog: Osteoarthritis of the hip or knee, allele T, OR 1.05, p=2E-10, PMID 30664745]	Moderate
FILIP1/ SENP6	rs9350591		C/C	chip	T (0 copy/s)	favourable	Locus 6q14: hip arthrosis [GWAS Catalog: Osteoarthritis (hip), allele T, OR 1.14, p=2E-13, PMID 30664745]	Moderate
TGFA	rs3771501		A/G	chip	A (1 copy/s)	unfavourable (light)	Transforming growth factor α: hip arthrosis [GWAS Catalog: Osteoarthritis, allele A, OR 1.05, p=4E-16, PMID 30664745]	Moderate

Hand arthrosis and signaling

Retinoic acid (ALDH1A2) and neuronal signaling of the hand joint.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ALDH1A2	rs3204689		G/C	chip	C (1 copy/s)	unfavourable (light)	Synthesis of retinoic acid in cartilage: the strongest locus of severe hand arthrosis [GWAS Catalog: Osteoarthritis (hand, severe), allele C, OR 1.46, p=1E-11, PMID 24728293]	Strong
ASTN2	rs4836732		T/T	chip	C (0 copy/s)	favourable	Locus ASTN2: hip arthrosis [GWAS Catalog: Osteoarthritis (hip), C allele, OR 1.09, p=4E-12, PMID 30374069]	Weak

Drop and crystals (linking to §30)

Renal and intestinal urate transporters.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ABCG2	rs2231142	Q141K	G/G	chip	T (0 copy/s)	favourable	Q141K: T allele reduces urate excretion; ~1.7× more gout per copy	Strong
SLC2A9	rs12498742		A/G	chip	A (1 copy/s)	unfavourable (light)	GLUT9 (urate renal resorption): the largest locus of uric acid; allele A rises urate (~0.3 mg/dL per copy) and droplet	Strong
SLC22A12/ NRXN2	rs2078267		C/T	chip	C (1 copy/s)	unfavourable (light)	URAT1 [GWAS Catalog: Gout, C allele, OR 1.28, p=2E-37, PMID 33832965]	Moderate

5. GWAS loci burden

Sum of loci GWAS Catalog ($p \leq 510-8$) by joint and droplet. Alleles are oriented towards more risk and the sum is compared with the reference population. The arthrosis of the hand has few loci.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Arthritis (any local)	588 (552)	517 of 1104	550	20.2	typical
Knee arthrosis	514 (476)	478 of 952	474	29.5	typical
Hip arthrosis	563 (524)	517 of 1048	521	50.4	typical
Arthritis of the hand and fingers	37 (37)	47 of 74	37	97.6	unfavourable
Drop	640 (585)	539 of 1170	585	34.4	typical

6. Favourable

- **Arthritis in any location:** percentile 15.8; loci load at 20.2. **Moderate.**
- **Knee:** loci load at 29.5; total joint replacement at 41.2; 83 score variants in 6.0 (44% coverage). **Moderate.**
- **Hip:** total joint replacement at 40.6; 70 score variants in 11.4 (34% coverage); typical loci load (50.4). **Weak to moderate.**
- **GDF5 rs143383 G/G, RUNX2 A/A, FILIP1 C/C and ASTN2 T/T:** without risk alleles. Moderate (GDF5).
- **Drop:** genomic score at 12.3, loci load at 34.4; **ABCG2 Q141K G/G. Moderate to strong.**

7. Unfavourable

- **Arthritis of the hand and fingers:** loci load at 97.6; ALDH1A2 G/C (1 allele, ~1.5× in severe form). **Moderate.**
- **MCF2L A/A** (2 alleles) and **DOT1L C/C** (without hip protective allele); GLT8D1 and TGFA with 1 allele. Weak (~1.05–1.2× each).
- **Knee and hip, typical genomic scores:** 56.1 e 38.9. **Moderate.**
- **Urate:** SLC2A9 (GLUT9) A/G and SLC22A12 (URAT1) C/T, 1 allele each; the other droplet scores are at 35.1 and 56.7. **Weak.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Remodelling of the extracellular matrix	Cartilage and subchondral bone (GDF5 G/G favourable; MCF2L, DOT1L and TGFA unfavourable); hand arthrosis with loci in 97.6	Unfavourable (mild)
Stiffness of the Extracellular Matrix	The cartilage hardens with age (cross links and AGEs) and loses the ability to damp	Conditional
Accumulation of Advanced Glycation Final Products (AGEs)	The AGEs accumulate in the collagen of the cartilage; controlled glycemia and low roasted diet slow the process	Conditional
Cell Senescence	Senescent chondrocytes (secretory phenotype) degrade cartilage; target of senolytics in trial	Conditional
Chronic Inflammation	Low-grade synovitis accelerates the arthrosis; the droplet is unlikely (ABCG2 G/G, genomic at percentile 12.3)	Favourable
Loss of muscle strength	Strong quadriceps is the main protector of the knee; strength training is already in the plan	Conditional
Fragility	Large joint arthrosis has typical or low genetic risk	Favourable

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Strength training focused on quadriceps, glutes and handgrip**, 2–3 times a week. It is the intervention with more evidence to prevent and treat knee and hand arthrosis. *Suggestion (not applied)*.
2. **Protecting joints from injuries**: heating, technique and progressive increase of the load in training. A rupture of ligament or meniscus is the major risk factor isolated from knee arthrosis. *Suggestion (not applied)*.
3. **Hand**: be attentive to nodules in the joints of the fingers and short morning pain or stiffness. If they arise, the first steps are hand exercises and topic NSAID. There is no proven pharmacological prevention. *Suggestion (not applied)*.
4. **Glucosamine and chondroitin**: weak and inconsistent evidence; not adding to the plan. *Suggestion (not applied)*.
5. **Drop**: the uric acid in routine analyzes arrives. The dapagliflozin already lowers the urate and the alcohol zero protects. Avoid sugary beverages with fructose. *Suggestion (not applied)*.
6. **Already in the plan and protectors**: Omega-3 3.2–4 g/day, vitamin D, daily exercise and controlled glycemia (less AGEs in cartilage). *Suggestion (not applied)*.

10. Limitations

- Knee and hip scores of 70–83 variants have 34–44% coverage; genomics prevail.
- The hand arthrosis has few loci (37); the high percentile is read as trend.
- The chip does not read the rare variants of dysplasias and chondrodysplasias (COL2A1, COMP and others).
- Arthritis depends mainly on age, injuries, mechanical load and muscle strength; genetics explains a modest part.
- The load of loci GWAS sums SNPs in linkage imbalance; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). The absent SNPs come from local imputation (Beagle 5.5, public panel of complete genomes to 30x), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the imputed VCF; in the palindromic SNPs the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed. The alleles published with protective effect were reversed to the sense of risk.
- **Polygenic scores:** GRCh38 files from PGS Catalog (10 scores: general arthrosis, knee and hip, prostheses and droplet). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510^{-8}$), 5 sets (2 342 loci), alleles oriented by the sign of the published effect, and empirical calibration in the same reference population.
- **Interpretation:** Boer 2021 and Tachmazidou 2019 (GWAS of the arthrosis), Styrkarsdottir 2014 (ALDH1A2), Miyamoto 2007 (GDF5), Wei 2024 and Lacaze 2022 (scores), Major 2018 (urate), GWAS Catalog and ClinVar.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Cartilage and joint shape	GDF5	rs143383	G/G	chip	T	0	complement (strand)	without the risk allele
Cartilage and joint shape	DOT1L	rs12982744	C/C	chip	C	2	strand + (palindromic; assumed +)	C/C: without the G allele, associated with more hip joint space
Cartilage and joint shape	SUPT3H/ RUNX2	rs10948172	A/A	chip	G	0	strand +	without the risk allele
Cartilage and joint shape	MCF2L	rs11842874	A/A	chip	A	2	strand +	2 risk allele copies
Cartilage and joint shape	GLT8D1	rs6976	C/T	chip	T	1	strand +	1 copy of risk allele
Cartilage and joint shape	FILIP1/ SENP6	rs9350591	C/C	chip	T	0	strand +	without the risk allele
Cartilage and joint shape	TGFA	rs3771501	A/G	chip	A	1	strand +	1 copy of risk allele
Hand arthrosis and signaling	ALDH1A2	rs3204689	G/C	chip	C	1	strand + (palindromic; assumed +)	1 copy of risk allele
Hand arthrosis and signaling	ASTN2	rs4836732	T/T	chip	C	0	strand +	without the risk allele
Drop and crystals (linking to §30)	ABCG2	rs2231142	G/G	chip	T	0	strand +	without the risk allele
Drop and crystals (linking to §30)	SLC2A9	rs12498742	A/G	chip	A	1	without data	A/G: 1 copy of allele A, which rises to uric acid
Drop and crystals (linking to §30)	SLC22A12/ NRXN2	rs2078267	C/T	chip	C	1	strand +	1 copy of risk allele